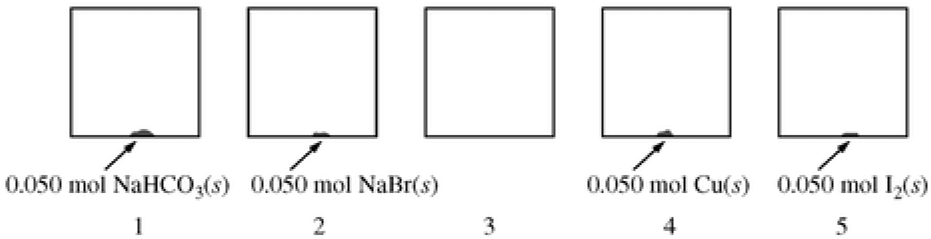
## Chapter 05 Part 2: Stoichiometry




At 27°C, five identical rigid 2.0 L vessels are filled with $N_2(g)$ and sealed. Four of the five vessels also contain a 0.050 mol sample of $NaHCO_3(s)$, $NaBr(s)$, $Cu(s)$, or $I_2(s)$, as shown in the diagram above. The volume taken up by the solids is negligible, and the initial pressure of $N_2(g)$ in each vessel is 720 mm Hg. All four vessels are heated to 127°C and allowed to reach a constant pressure.

**1.** At 127°C, the entire sample of $I_2$ is observed to have vaporized. How does the mass of vessel 5 at 127°C compare to its mass at 27°C?

(A) The mass is less, since the $I_2$ is in the vapor phase.

(B) The mass is the same, since the number of each type of atom in the vessel is constant.

(C) The mass is greater, since the $I_2$ will react with $N_2$ to form $NI_3$ , which has a greater molar mass.

(D) The mass is greater, since the pressure is greater and the particles have a higher average kinetic energy.

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**2.**

$$Fe(s) + 2\ HCl(aq) \rightarrow FeCl_2(aq) + H_2(g)$$

When a student adds 30.0 mL of 1.00 *M* HCl to 0.56 g of powdered Fe, a reaction occurs according to the equation above. When the reaction is complete at 273 K and 1.0 atm, which of the following is true?

(A) HCl is in excess, and 0.100 mol of HCl remains unreacted.

(B) HCl is in excess, and 0.020 mol of HCl remains unreacted.

(C) 0.015 mol of $FeCl_2$ has been produced.

(D) 0.22 L of $H_2$ has been produced.

**3.** A 20.0–milliliter sample of 0.200–molar $K_2CO_3$ solution is added to 30.0 milliliters of 0.400–molar $Ba(NO_3)_2$ solution. Barium carbonate precipitates. The concentration of barium ion, $Ba^{2+}$, in solution after reaction is

(A) 0.150 M

(B) 0.160 M

(C) 0.200 M

(D) 0.240 M

(E) 0.267 M

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A $0.50\ \mathrm{g}$ sample of $\mathrm{Mg}(s)$ was placed in a solution of $\mathrm{HCl}(aq)$, where it reacted completely.

**4.** In another experiment, a small piece of $\mathrm{Mg}(s)$ is weighed, then placed in a flask containing excess $1\ M\ \mathrm{HCl}(aq)$. The student wants to determine number of moles of gas produced. Which of the following is the best way to conduct the experiment for accurate data collection?

(A) Conducting the experiment at different temperatures to see which generates the most gas

(B) Completing the entire reaction in a large Erlenmeyer flask of known volume to measure the volume of the gas collected

(C) Collecting the gas in a eudiometer tube and measuring the volume of the gas collected

(D) Conducting the reaction in a graduated cylinder and measuring the volume of the gas collected

Reaction 1: $CaC_2(s) + 2\ H_2O(l) \rightarrow C_2H_2(g) + Ca(OH)_2(s)$

Reaction 2: $NaOCl(aq) + 2\ HCl(aq) \rightarrow Cl_2(g) + NaCl(aq) + H_2O(l)$

Reaction 3: $C_2H_2(g) + Cl_2(g) \rightarrow C_2H_2Cl_2(g)$

**5.** Reaction 2 occurs when an excess of 6 *M* HCl(*aq*) solution is added to 100. mL of NaOCL(*aq*) of unknown concentration. If the reaction goes to completion and 0.010 mol of $Cl_2(g)$ is produced, then what was the molarity of the NaOCL(*aq*) solution?

(A) 0.0010 *M*

(B) 0.010 *M*

(C) 0.10 *M*

(D) 1.0 *M*

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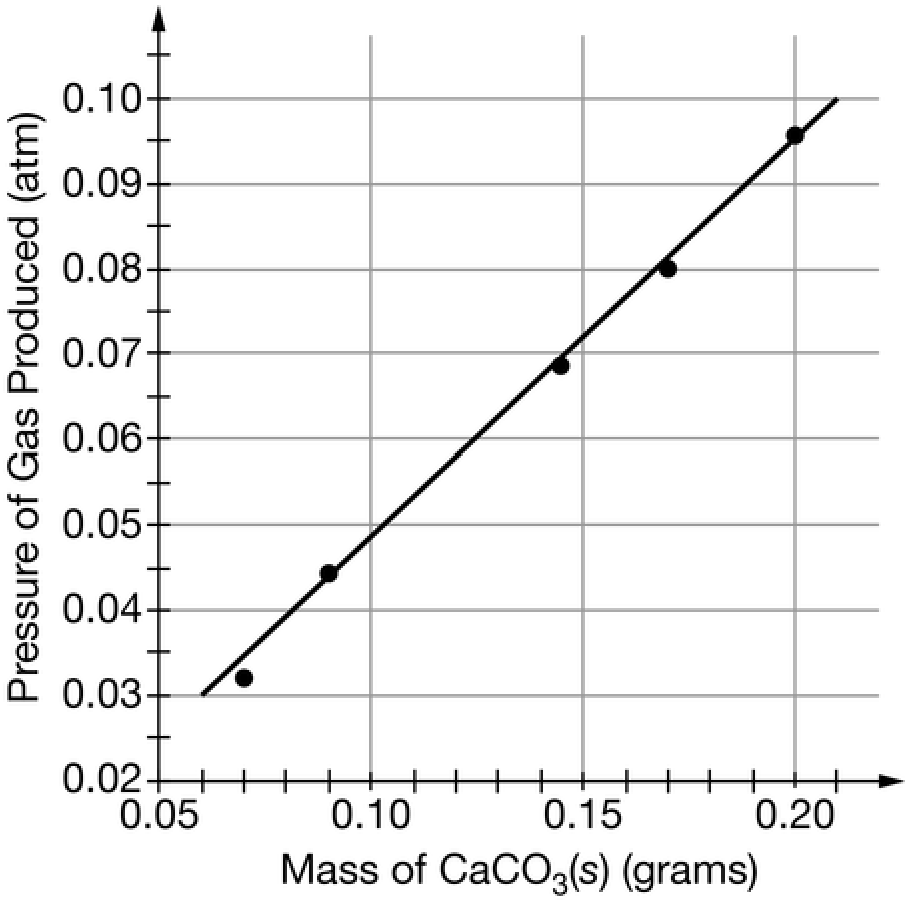
An experiment is performed to measure the mass percent of $CaCO_3(s)$ in eggshells. Five different samples of $CaCO_3(s)$ of known mass react with an excess of $2.0\ M\ HCl(aq)$ in identical sealed, rigid reaction vessels. The pressure of the gas produced is measured with a pressure gauge attached to the reaction vessel. Since the reaction is exothermic, the reaction system is cooled to its original temperature before the pressure is recorded. The experimental data are used to create the calibration line below.




The experiment is repeated with an eggshell sample, and the experimental data are recorded in the table below.

| | |
|---|---|
| Mass of eggshell sample | $0.200$ g |
| Pressure prior to reaction | $0.800$ atm |
| Pressure at completion of reaction | $0.870$ atm |

**6.** Another sample of eggshell reacts completely with $4.0$ mL of an $HCl(aq)$ solution of unknown concentration. If the reaction produced $0.095$ atm of gas, the concentration of the $HCl(aq)$ solution was at least

(A) 0.0020 $M$
(B) 0.050 $M$
(C) 0.50 $M$
(D) 1.0 $M$

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$$CH_3OH(g) \rightarrow CO(g) + 2\,H_2(g) \qquad \Delta H^\circ = +91 \text{ kJ/mol}_{rxn}$$

The reaction represented above goes essentially to completion. The reaction takes place in a rigid, insulated vessel that is initially at 600 K

**7.** A sample of $CH_3OH(g)$ is placed in the previously evacuated vessel with a pressure of $P_1$ at 600 K. What is the final pressure in the vessel after the reaction is complete and the contents of the vessel are returned to 600 K?

(A) $P_1 / 9$
(B) $P_1 / 3$
(C) $P_1$
(D) $3P_1$

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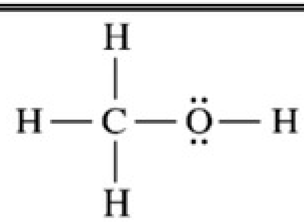
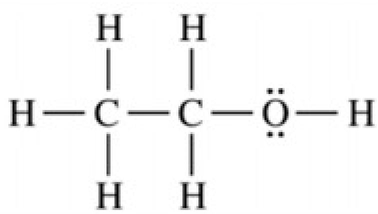
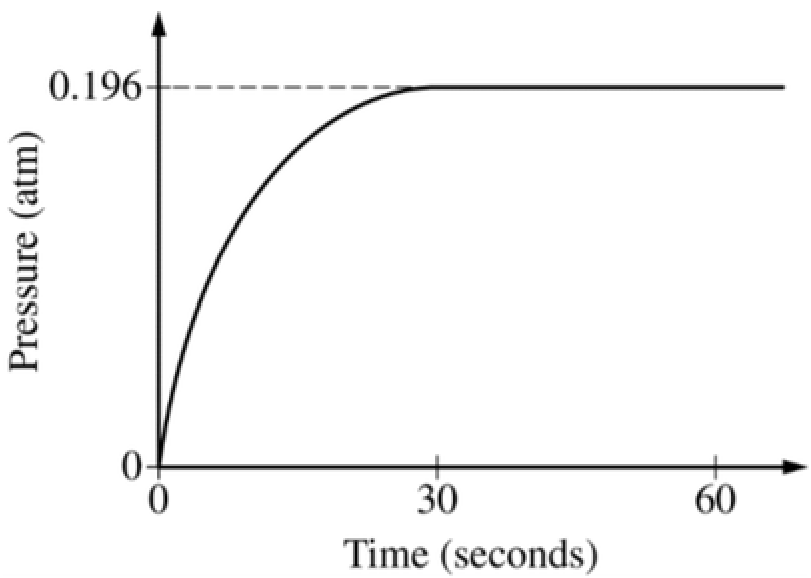
| Substance | Lewis Diagram | Boiling Point |
|---|---|---|
| $CH_3OH$ |  | 338 K |
| $C_2H_5OH$ |  | 351 K |

Equimolar samples of $CH_3OH(l)$ and $C_2H_5OH(l)$ are placed in separate, previously evacuated, rigid 2.0 L vessels. Each vessel is attached to a pressure gauge, and the temperatures are kept at 300 K. In both vessels, liquid is observed to remain present at the bottom of the container at all times. The change in pressure inside the vessel containing $CH_3OH(l)$ is shown below.




**8.** The temperature of the $CH_3OH$ is increased from 300 K to 400 K to vaporize all the liquid, which increases the pressure in the vessel to 0.30 atm. The experiment is repeated under identical conditions but this time using half the mass of $CH_3OH$ that was used originally. What will be the pressure in the vessel at 400 K ?

(A) 0.15 atm
(B) 0.30 atm
(C) 0.40 atm
(D) 0.60 atm

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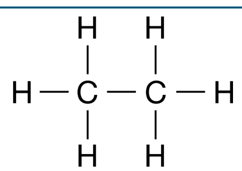
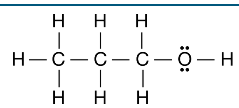
**9.** $2\,F_2(g) + 2\,NaOH(aq) \rightarrow OF_2(g) + 2\,NaF(aq) + H_2O(l)$

A $2\ mol$ sample of $F_2(g)$ reacts with excess $NaOH(aq)$ according to the equation above. If the reaction is repeated with excess $NaOH(aq)$ but with $1\ mol$ of $F_2(g)$, which of the following is correct?

(A) The amount of $OF_2(g)$ produced is doubled.
(B) The amount of $OF_2(g)$ produced is halved.
(C) The amount of $NaF(aq)$ produced remains the same.
(D) The amount of $NaF(aq)$ produced is doubled.

**10.** If equal masses of the following compounds undergo complete combustion, which will yield the greatest mass of $CO_2$?

(A) Benzene, $C_6H_6$
(B) Cyclohexane, $C_6H_{12}$
(C) Glucose, $C_6H_{12}O_6$
(D) Methane, $CH_4$

| Compound | Molar Mass | Balanced Equation for the Combustion of One Mole of the Compound | $\Delta H^\circ_{comb}$ $(kJ/mol_{rxn})$ |
|---|---|---|---|
|  Ethane | $30.\,g/mol$ | $C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2\,CO_2(g) + 3\,H_2O(l)$ | $-1560$ |
|  Propanol | $60.\,g/mol$ | $C_3H_8O(l) + \frac{9}{2}O_2(g) \rightarrow 3\,CO_2(g) + 4\,H_2O(l)$ | $-2020$ |
| Unknown | ? | $? +\ 5\,O_2(g) \rightarrow 4\,CO_2(g) + 4\,H_2O(l)$ | $-2240$ |

**11.** Which of the following is the molecular formula of the unknown compound?

(A) $C_4H_8$
(B) $C_4H_8O$
(C) $C_4H_4O_2$
(D) $C_4H_8O_2$

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**12.** $5\ Fe^{2+} + MnO_4^{-} + 8\ H^{+} \rightleftarrows 5\ Fe^{3+} + Mn^{2+} + 4\ H_2O$

In a titration experiment based on the equation above, 25.0 milliliters of an acidified $Fe^{2+}$ solution requires 14.0 milliliters of standard 0.050–molar $MnO_4^{-}$ solution to reach the equivalence point. The concentration of $Fe^{2+}$ in the original solution is

(A) 0.0010 *M*

(B) 0.0056 *M*

(C) 0.0028 *M*

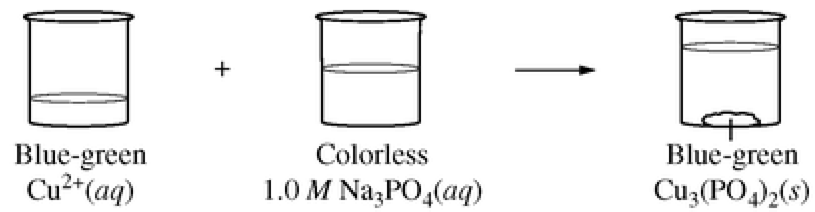
(D) 0.0090 *M*

(E) 0.14 *M*

## Chapter 05 Part 2: Stoichiometry

A group of students was asked to recover Cu($s$) from a blue-green aqueous solution containing an unknown concentration of $Cu^{2+}$($aq$) . The students took a 100.0 mL sample of the solution and added an excess of 1.0 $M$ $Na_3PO_4$($aq$), causing the $Cu^{2+}$($aq$) to precipitate as $Cu_3(PO_4)_2$($s$), as shown in step 1 below.

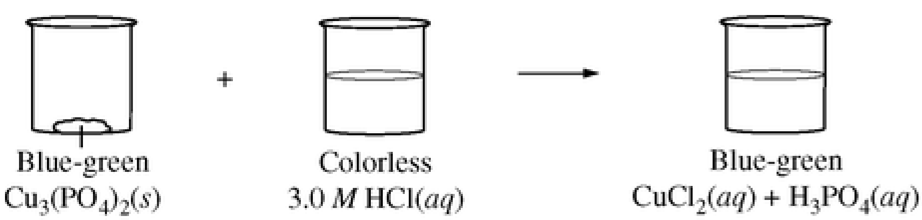
Step 1




$$3\ Cu^{2+}(aq) + 2\ Na_3PO_4\ (aq) \longrightarrow Cu_3(PO_4)_2(s) + 6\ Na^+(aq)$$

The $Cu_3(PO_4)_2$($s$) was filtered, dried, and weighed. Then the $Cu_3(PO_4)_2$($s$) was dissolved in a 3.0 $M$ HCl($aq$) solution, as shown in step 2 below.

Step 2

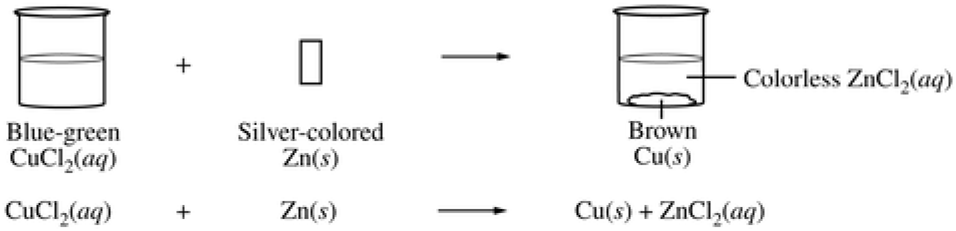



$$Cu_3(PO_4)_2(s) + 6\ HCl(aq) \longrightarrow 3\ CuCl_2(aq) + 2\ H_3PO_4(aq)$$

The students added a strip of Zn($s$) to the solution to recover the Cu($s$), as shown in step 3 below.

Step 3

## Chapter 05 Part 2: Stoichiometry




Finally, the Cu($s$) was filtered, dried, and weighed.

**13.**

If 3.8 g of $Cu_3(PO_4)_2$($s$) was recovered from step 1, what was the approximate [$Cu^{2+}$] in the original solution? (The molar mass of $Cu_3(PO_4)_2$ is 381 g/mol.)

(A) 0.10 *M*

(B) 0.30 *M*

(C) 1.0 *M*

(D) 3.0 *M*

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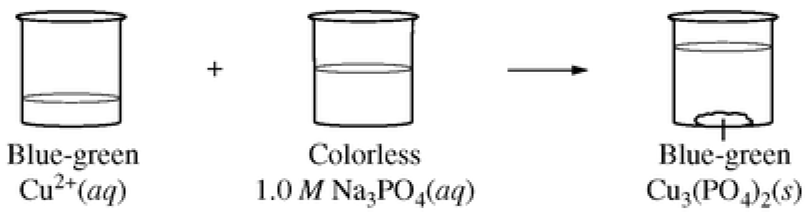
**14.**

A group of students was asked to recover Cu(*s*) from a blue-green aqueous solution containing an unknown concentration of $Cu^{2+}(aq)$ . The students took a 100.0 mL sample of the solution and added an excess of 1.0 *M* $Na_3PO_4(aq)$, causing the $Cu^{2+}(aq)$ to precipitate as $Cu_3(PO_4)_2(s)$, as shown in step 1 below.

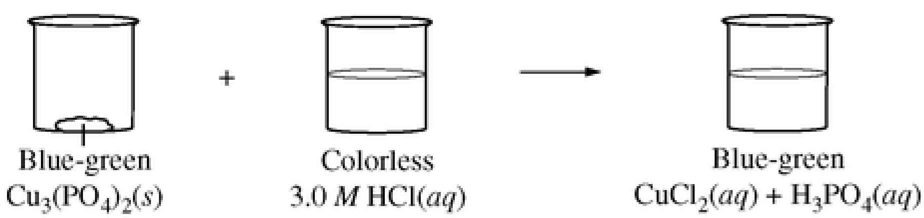
Step 1




$$3\,Cu^{2+}(aq) + 2\,Na_3PO_4(aq) \longrightarrow Cu_3(PO_4)_2(s) + 6\,Na^+(aq)$$

The $Cu_3(PO_4)_2(s)$ was filtered, dried, and weighed. Then the $Cu_3(PO_4)_2(s)$ was dissolved in a 3.0 *M* HCl(*aq*) solution, as shown in step 2 below.

Step 2

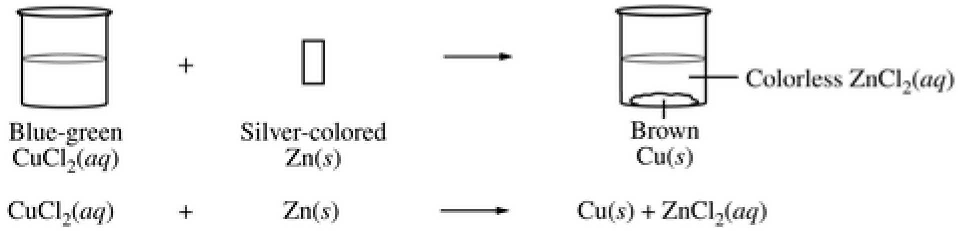



$$Cu_3(PO_4)_2(s) + 6\,HCl(aq) \longrightarrow 3\,CuCl_2(aq) + 2\,H_3PO_4(aq)$$

The students added a strip of Zn(*s*) to the solution to recover the Cu(*s*), as shown in step 3 below.

Step 3

## Chapter 05 Part 2: Stoichiometry




Finally, the Cu(*s*) was filtered, dried, and weighed.

If 4.6 g of $Cu_3(PO_4)_2(s)$ was recovered from step 1, what was the approximate $[Cu^{2+}]$ in the original solution? (The molar mass of $Cu_3(PO_4)_2$ is 381 g/mol.)

(A) 0.040 *M*
(B) 0.12 *M*
(C) 0.36 *M*
(D) 1.5 *M*

**15.** $\mathrm{MgCl_2}(aq) + 2\ \mathrm{NaOH}(aq) \rightarrow 2\ \mathrm{NaCl}(aq) + \mathrm{Mg(OH)_2}(s)$

A $100\ \mathrm{mL}$ sample of $0.1\ M\ \mathrm{MgCl_2}(aq)$ and a $100\ \mathrm{mL}$ sample of $0.2\ M\ \mathrm{NaOH}(aq)$ were combined, and $\mathrm{Mg(OH)_2}(s)$ precipitated, as shown by the equation above. If the experiment is repeated using solutions of the same molarity, which of the following changes in volume will double the amount of $\mathrm{Mg(OH)_2}(s)$ produced?

(A) Using the same volume of $\mathrm{MgCl_2}(aq)$ but twice the volume of $\mathrm{NaOH}(aq)$
(B) Using twice the volume of $\mathrm{MgCl_2}(aq)$ but half the volume of $\mathrm{NaOH}(aq)$
(C) Using twice the volume of $\mathrm{MgCl_2}(aq)$ but the same volume of $\mathrm{NaOH}(aq)$
(D) Using twice the volume of $\mathrm{MgCl_2}(aq)$ and twice the volume of $\mathrm{NaOH}(aq)$

**16.** $$2\ SO_2(g) + O_2(g) \rightarrow 2\ SO_3(g)$$

A mixture of gases containing 0.20 mol of $SO_2$ and 0.20 mol of $O_2$ in a 4.0 L flask reacts to form $SO_3$. If the temperature is 25ºC, what is the pressure in the flask after reaction is complete?

(A) $\frac{0.4(0.082)(298)}{4} atm$
(B) $\frac{0.3(0.082)(298)}{4} atm$
(C) $\frac{0.2(0.082)(298)}{4} atm$
(D) $\frac{0.2(0.082)(25)}{4} atm$
(E) $\frac{0.3(0.082)(25)}{4} atm$

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$Li_3N(s) + 2\ H_2(g) \rightleftarrows LiNH_2(s) + 2\ LiH(s)$

$\Delta H° = -192\ kJ/mol_{rxn}$

Because pure $H_2$ is a hazardous substance, safer and more cost effective techniques to store it as a solid for shipping purposes have been developed. One such method is the reaction represented above, which occurs at 200°C.

**17.** When 70. g of $Li_3N(s)$ (molar mass 35 g/mol) reacts with excess $H_2(g)$, 8.0 g of $LiH(s)$ is produced. The percent yield is closest to

(A) 17%

(B) 25%

(C) 50.%

(D) 100%

$$2\ H_2O_2(aq) \rightarrow 2\ H_2O(l) + O_2(g) \qquad \Delta H° = -196\ kJ/mol_{rxn}$$

The decomposition of $H_2O_2(aq)$ is represented by the equation above. A student monitored the decomposition of a 1.0 L sample of $H_2O_2(aq)$ at a constant temperature of 300. K and recorded the concentration of $H_2O_2$ as a function of time. The results are given in the table below.

| Time (s) | $[H_2O_2]$ |
|---|---|
| 0 | 2.7 |
| 200. | 2.1 |
| 400. | 1.7 |
| 600. | 1.4 |

**18.** The $O_2(g)$ produced from the decomposition of the 1.0 L sample of $H_2O_2(aq)$ is collected in a previously evacuated 10.0 L flask at 300. K. What is the approximate pressure in the flask after 400. s? (For estimation purposes, assume that 1.0 mole of gas in 1.0 L exerts a pressure of 24 atm at 300. K.)

(A) 1.2 atm
(B) 2.4 atm
(C) 12 atm
(D) 24 atm

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$$K(s) + \tfrac{1}{2} Cl_2(g) \rightarrow KCl(s) \qquad \Delta H° = -437 kJ/mol_{rxn}$$

The elements K and Cl react directly to form the compound KCl according to the equation above. Refer to the information above and the table below to answer the questions that follow.

| Process | $\Delta H°$ ($kJ/mol_{rxn}$) |
|---|---|
| $K(s) \rightarrow K(g)$ | $v$ |
| $K(g) \rightarrow K^+(g) + e^-$ | $w$ |
| $Cl_2(g) \rightarrow 2\,Cl(g)$ | $x$ |
| $Cl(g) + e^- \rightarrow Cl^-(g)$ | $y$ |
| $K^+(g) + Cl^-(g) \rightarrow KCl(s)$ | $z$ |

**19.** What remains in the reaction vessel after equal masses of $K(s)$ and $Cl_2(g)$ have reacted until either one or both of the reactants have been completely consumed?

(A) KCl only
(B) KCl and K only
(C) KCl and $Cl_2$ only
(D) KCl, K, and $Cl_2$

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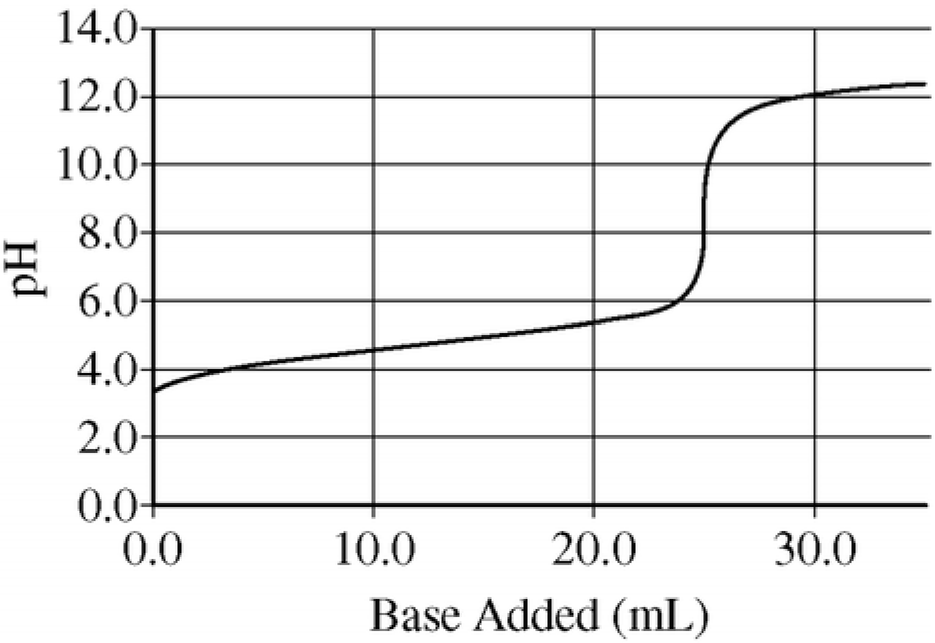
A 0.35 g sample of Li($s$) is placed in an Erlenmeyer flask containing 100 mL of water at 25°C. A balloon is placed over the mouth of the flask to collect the hydrogen gas that is generated.

After all of the Li($s$) has reacted with $H_2O(l)$, the solution in the flask is added to a clean, dry buret and used to titrate an aqueous solution of a monoprotic acid. The pH curve for this titration is shown in the diagram below




**20.** What will be the effect on the amount of gas produced if the experiment is repeated using 0.35 g of K($s$) instead of 0.35 g of Li($s$) ?

(A) No gas will be produced when K($s$) is used.

(B) Some gas will be produced but less than the amount of gas produced with Li($s$).

(C) Equal quantities of gas will be produced with the two metals.

(D) More gas will be produced with K($s$) than with Li($s$).

## Chapter 05 Part 2: Stoichiometry

$4\ \mathrm{FeS}(s)\ +\ 7\ \mathrm{O_2}(g)\ \rightarrow\ 2\ \mathrm{Fe_2O_3}(s)\ +\ 4\ \mathrm{SO_2}(g)$

Due to the presence of $\mathrm{FeS}(s)$ as an impurity, the combustion of some types of coal results in the formation of $\mathrm{SO_2}(g)$, as represented by the equation above. Also, $\mathrm{SO_2}(g)$ can react with $\mathrm{O_3}(g)$ to form $\mathrm{SO_3}(g)$, as represented by the equation below.

$\mathrm{SO_2}(g)\ +\ \mathrm{O_3}(g)\ \rightleftarrows\ \mathrm{SO_3}(g)\ +\ \mathrm{O_2}(g)$

$\Delta H^\circ_{298} = -242\ \mathrm{kJ/mol}_{rxn}\ ;\ \Delta S^\circ_{298} = -25\ \mathrm{J/(K \cdot mol}_{rxn})$

**21.** Which of the following statements is correct when $2.0\ \mathrm{mol}$ of $\mathrm{FeS}(s)$ reacts with $4.0\ \mathrm{mol}$ of $\mathrm{O_2}(g)$ ?

(A) $\mathrm{O_2}(g)$ is the limiting reactant and $2.0\ \mathrm{mol}$ of $\mathrm{SO_2}(g)$ is formed.

(B) $\mathrm{O_2}(g)$ is the limiting reactant and $4.0\ \mathrm{mol}$ of $\mathrm{SO_2}(g)$ is formed.

(C) $\mathrm{FeS}(s)$ is the limiting reactant and $4.0\ \mathrm{mol}$ of $\mathrm{SO_2}(g)$ is formed.

(D) $\mathrm{FeS}(s)$ is the limiting reactant and $0.5\ \mathrm{mol}$ of $\mathrm{O_2}(g)$ remains unreacted.

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**22.** If 0.40 mol of $H_2$ and 0.15 mol of $O_2$ were to react as completely as possible to produce $H_2O$, what mass of reactant would remain?

(A) 0.20 g of $H_2$

(B) 0.40 g of $H_2$

(C) 3.2 g of $O_2$

(D) 4.0 g of $O_2$

(E) 4.4 g of $O_2$

**23.**

$$2\ N_2H_4(g) + N_2O_4(g) \rightarrow 3\ N_2(g) + 4\ H_2O(g)$$

When 8.0 g of $N_2H_4$ (32 g mol$^{-1}$) and 92 g of $N_2O_4$ (92 g mol$^{-1}$) are mixed together and react according to the equation above, what is the maximum mass of $H_2O$ that can be produced?

(A) 9.0 g

(B) 18 g

(C) 36 g

(D) 72 g

(E) 144 g

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**24.** $$2\ Al(s) + 6\ HCl(aq) \rightarrow 2\ AlCl_3(aq) + 3\ H_2(g)$$

According to the reaction represented above, about how many grams of aluminum (atomic mass 27 g) are necessary to produce 0.50 mol of hydrogen gas at 25ºC and 1.00 atm?

(A) 1.0 g
(B) 9.0 g
(C) 14 g
(D) 27 g
(E) 56 g

**25.** What mass of Au is produced when 0.0500 mol of $Au_2S_3$ is reduced completely with excess $H_2$?

(A) 9.85 g
(B) 19.7 g
(C) 24.5 g
(D) 39.4 g
(E) 48.9 g

**26.** Step 1: $4\ FeS_2(s) + 11\ O_2(g) \rightarrow 2\ Fe_2O_3(s) + 8\ SO_2(g)$
Step 2: $Fe_2O_3(s) + 3\ CO(g) \rightarrow 3\ CO_2(g) + 2\ Fe(s)$

Extraction of $Fe(s)$ from pyrite, $FeS_2$, is a two-step process, as represented above. The maximum amount of $Fe(s)$ that could be extracted from $120$ grams of $FeS_2$ (molar mass $120\ g/mol$) is approximately

(A) 28 grams
(B) 56 grams
(C) 84 grams
(D) 112 grams

**27.** $$H_2 + F_2 \rightarrow 2\ HF$$

In the reaction represented above, what mass of HF is produced by the reaction of 3.0 x $10^{23}$ molecules of $H_2$ with excess $F_2$ ? (Assume the reaction goes to completion.)

(A) 1.0 g
(B) 4.0 g
(C) 10. g
(D) 20. g
(E) 40. g

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**28.** $CO_2(g) + 2\ LiOH(s) \rightarrow Li_2CO_3(aq) + H_2O(l)$

In a one-person spacecraft, an astronaut exhales $967\ g$ of $CO_2(g)$ (molar mass $44.01\ g/mol$) per day. To prevent the buildup of $CO_2(g)$ in the spacecraft, a device containing $LiOH(s)$ is used to remove the $CO_2(g)$, as represented by the equation above. What mass of $LiOH(s)$ (molar mass $23.95\ g/mol$) is needed to react with all of the $CO_2(g)$ produced by an astronaut in one day

(A) $43.9\ g$
(B) $263\ g$
(C) $526\ g$
(D) $1050\ g$

**29.**

$CO_2(g) + 2\ LiOH(s) \rightarrow Li_2CO_3(aq) + H_2O(l)$

In a one-person spacecraft, an astronaut exhales $880\ g$ of $CO_2(g)$ (molar mass $44\ g/mol$) per day. To prevent the buildup of $CO_2(g)$ in the spacecraft, a device containing $LiOH(s)$ is used to remove the $CO_2(g)$, as represented by the equation above. What mass of $LiOH(s)$ (molar mass $24\ g/mol$) is needed to react with all of the $CO_2(g)$ produced by an astronaut in one day?

(A) $40.\ g$
(B) $240\ g$
(C) $480\ g$
(D) $960\ g$

**30.** $C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O$

The reaction between $C_6H_{12}O_6$ and $O_2$ is represented by the balanced equation above. In an experiment, $0.30\ mol$ of $CO_2$ was produced from the reaction of $0.05\ mol$ of $C_6H_{12}O_6$ with excess $O_2$. The reaction was repeated at the same temperature and in the same container, but this time $0.60\ mol$ of $CO_2$ was produced. Which of the following must be true?

(A) The initial amount of $C_6H_{12}O_6$ in the container must have been $0.10\ mol$.
(B) Exactly $0.30\ mol$ of $C_6H_{12}O_6$ must have reacted because $C$ atoms were conserved.
(C) Exactly $0.40\ mol$ of $O_2$ of must have reacted because the temperature and container volume are the same.
(D) More than $0.60\ mol$ of $O_2$ must have reacted because it was present in excess.

**31.**

$$C_3H_8(g) + 4\ Cl_2(g) \rightarrow C_3H_4Cl_4(g) + 4\ HCl(g)$$

A 6.0 mol sample of $C_3H_8(g)$ and a 20. mol sample of $Cl_2(g)$ are placed in a previously evacuated vessel, where they react according to the equation above. After one of the reactants has been totally consumed, how many moles of $HCl(g)$ have been produced?

## Chapter 05 Part 2: Stoichiometry

(A) 4.0 mol
(B) 8.0 mol
(C) 20. mol
(D) 24 mol

**32.** What is the maximum number of moles of $Al_2O_3$ that can be produced by the reaction of 0.40 mol of Al with 0.40 mol of $O_2$ ?

(A) 0.10 mol
(B) 0.20 mol
(C) 0.27 mol
(D) 0.33 mol
(E) 0.40 mol

**33.**

$$2\ MnO_4^-(aq) + 5\ C_2O_4^{2-}(aq) + 16\ H^+(aq) \rightarrow 2\ Mn^{2+}(aq) + 10\ CO_2(g) + 8\ H_2O(l)$$

Permanganate and oxalate ions react in an acidified solution according to the balanced equation above. How many moles of $CO_2(g)$ are produced when 20. mL of acidified 0.20 *M* $KMnO_4$ solution is added to 50. mL of 0.10 *M* $Na_2C_2O_4$ solution?

(A) 0.0040 mol
(B) 0.0050 mol
(C) 0.0090 mol
(D) 0.010 mol
(E) 0.020 mol

**34.** A 1.0 L sample of an aqueous solution contains 0.10 mol of NaCl and 0.10 mol of $CaCl_2$. What is the minimum number of moles of $AgNO_3$ that must be added to the solution in order to precipitate all of the $Cl^-$ as $AgCl(s)$ ? (Assume that AgCl is insoluble.)

(A) 0.10 mol
(B) 0.20 mol
(C) 0.30 mol
(D) 0.40 mol
(E) 0.60 mol

**35.** $C_3H_8(g) + 5\ O_2(g) \rightarrow 3\ CO_2(g) + 4\ H_2O(l)$

In the reaction represented above, what is the total number of moles of reactants consumed when 1.00 mole of $CO_2(g)$ is produced?

(A) 0.33 mol
(B) 1.33 mol
(C) 1.50 mol
(D) 2.00 mol
(E) 6.00 mol

**36.**

$$2\ H_2O(l) + 4\ MnO_4^-(aq) + 3\ ClO_2^-(aq) \rightarrow 4\ MnO_2(s) + 3\ ClO_4^-(aq) + 4\ OH^-(aq)$$

According to the balanced equation above, how many moles of $ClO_2^-(aq)$ are needed to react completely with 20. mL of 0.20 $M$ $KMnO_4$ solution?

(A) 0.0030 mol
(B) 0.0053 mol
(C) 0.0075 mol
(D) 0.013 mol
(E) 0.030 mol

**37.** $10\ HI + 2\ KMnO_4 + 3\ H_2SO_4 \rightarrow 5\ I_2 + 2\ MnSO_4 + K_2SO_4 + 8\ H_2O$

According to the balanced equation above, how many moles of HI would be necessary to produce 2.5 mol of $I_2$, starting with 4.0 mol of $KMnO_4$ and 3.0 mol of $H_2SO_4$ ?

(A) 20.
(B) 10.
(C) 8.0
(D) 5.0
(E) 2.5

**38.**

$$2\ KClO_3\ (s) \xrightarrow[MnO_2]{\Delta} 2\ KCl\ (s) + 3\ O_2\ (g)$$

According to the equation above, how many moles of potassium chlorate, $KClO_3$, must be decomposed to generate 1.0 L of $O_2$ gas at standard temperature and pressure?

(A) $\frac{1}{3}\left(\frac{1}{22.4}\right)$ mol
(B) $\frac{1}{2}\left(\frac{1}{22.4}\right)$ mol
(C) $\frac{2}{3}\left(\frac{1}{22.4}\right)$ mol
(D) $\frac{3}{2}\left(\frac{1}{22.4}\right)$ mol
(E) $2\left(\frac{1}{22.4}\right)$ mol

## Chapter 05 Part 2: Stoichiometry

**39.** $3\ Ag(s) + 4\ HNO_3 \rightarrow 3\ AgNO_3 + NO(g) + 2\ H_2O$

The reaction of silver metal and dilute nitric acid proceeds according to the equation above. If 0.10 mole of powdered silver is added to 10. milliliters of 6.0–molar nitric acid, the number of moles of NO gas that can be formed is

(A) 0.015 mole
(B) 0.020 mole
(C) 0.030 mole
(D) 0.045
(E) 0.090

---

The following balanced equation represents the decomposition of hydrogen peroxide, $H_2O_2$.

$2\ H_2O_2(aq) \rightarrow 2\ H_2O(l) + O_2(g)$

A student carries out an experiment in which a sample of $H_2O_2(aq)$ decomposes in the presence of $MnO_2(s)$. The student measures the volume of $O_2(g)$ produced by the reaction at regular intervals while the temperature is held constant. The student's data are given in the following table.

| Time (s) | Volume of $O_2(g)$ Collected $(mL)$ |
|---|---|
| 0 | 0 |
| 60 | 42 |
| 120 | 64 |
| 180 | 76 |
| 240 | 82 |
| 300 | 86 |
| 360 | 88 |
| 420 | 89 |

**40.** If $6.0 \times 10^{-3}$ mol of $H_2O_2$ was consumed in the experiment, how many total moles of products were produced?

(A) $6.0 \times 10^{-3}$ mol
(B) $9.0 \times 10^{-3}$ mol
(C) $12 \times 10^{-3}$ mol
(D) $18 \times 10^{-3}$ mol

## Chapter 05 Part 2: Stoichiometry

**41.** When 100 mL of 1.0 *M* $Na_3PO_4$ is mixed with 100 mL of 1.0 *M* $AgNO_3$, a yellow precipitate forms and $[Ag^+]$ becomes negligibly small. Which of the following is a correct listing of the ions remaining in solution in order of increasing concentration?

(A) $[PO_4^{3-}] < [NO_3^-] < [Na^+]$
(B) $[PO_4^{3-}] < [Na^+] < [NO_3^-]$
(C) $[NO_3^-] < [PO_4^{3-}] < [Na^+]$
(D) $[Na^+] < [NO_3^-] < [PO_4^{3-}]$
(E) $[Na^+] < [PO_4^{3-}] < [NO_3^-]$

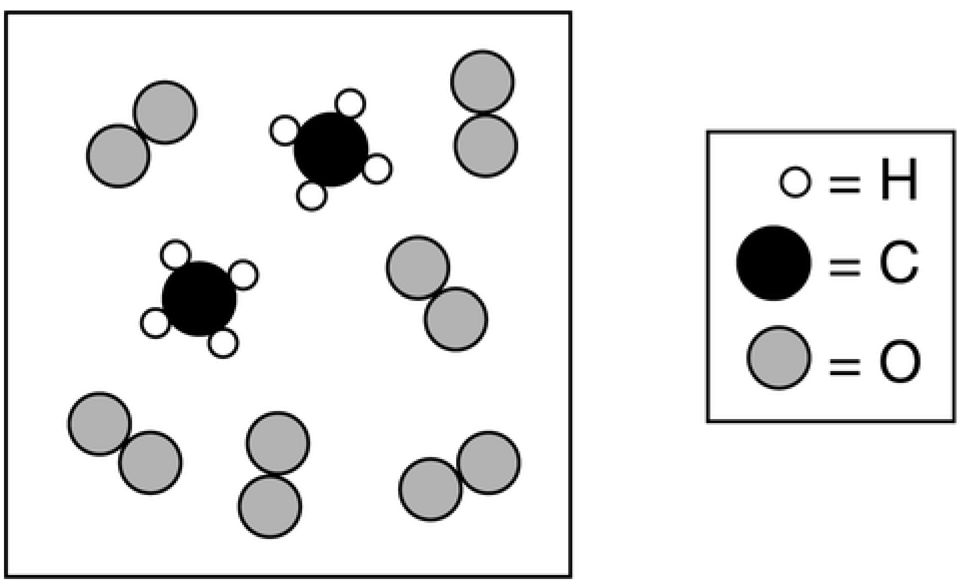
**42.**

$$MnO_4^- + 5\,Fe^{2+} + 8\,H^+ \rightarrow Mn^{2+} + 5\,Fe^{3+} + 4\,H_2O$$

In the reaction represented above, the number of $MnO_4^-$ ions that react must be equal to which of the following?

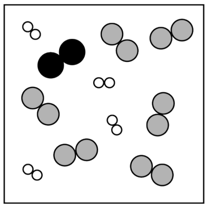
(A) One-fifth the number of $Fe^{2+}$ ions that are consumed
(B) Eight times the number of $H^+$ ions that are consumed
(C) Five times the number of $Fe^{3+}$ ions that are produced
(D) One-half the number of $H_2O$ molecules that are produced

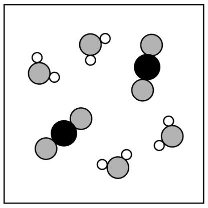
**43.**

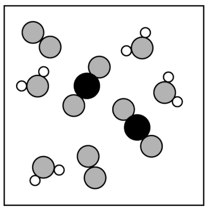


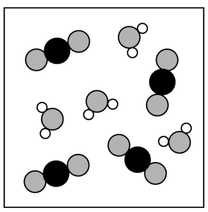

The reactants represented above are placed in a vessel and a reaction occurs. Which of the following best represents the contents of the vessel after the reaction has proceeded as completely as possible?

## Chapter 05 Part 2: Stoichiometry

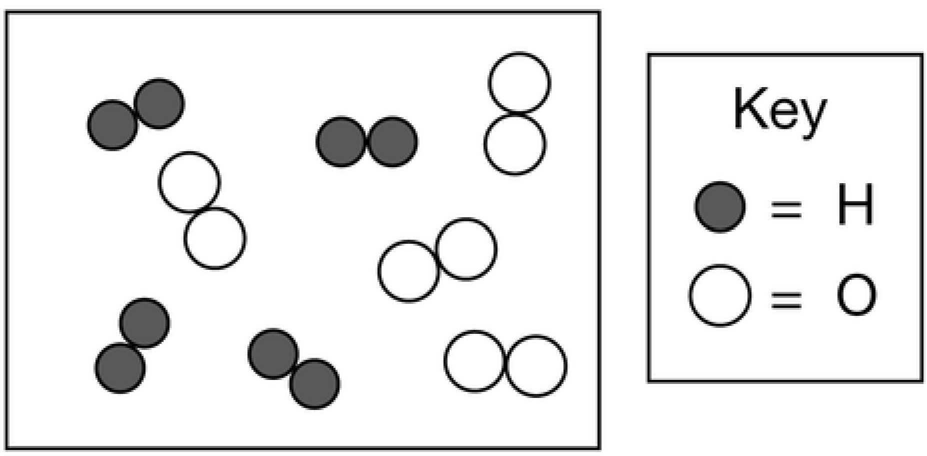
(A) 

(B) 

(C) 

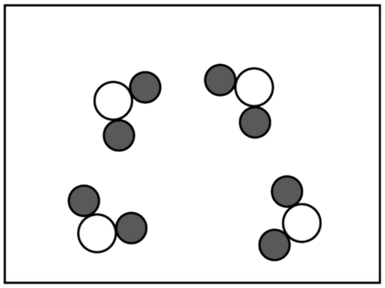
(D) 

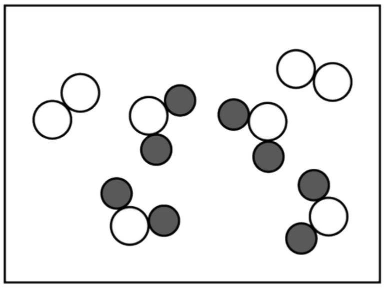
**44.**

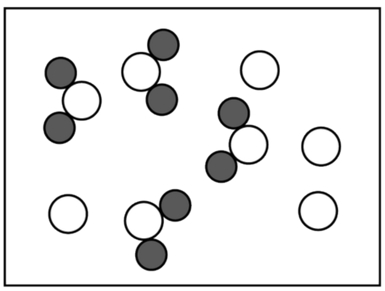


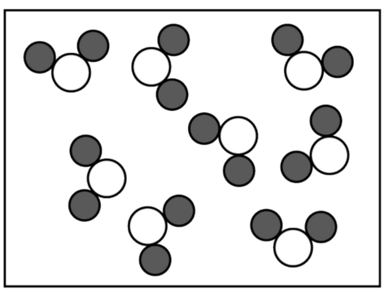

A mixture of $H_2(g)$ and $O_2(g)$ is placed in a container as represented above. The $H_2(g)$ and $O_2(g)$ react to form $H_2O(g)$. Which of the following best represents the container after the reaction has gone to completion?

## Chapter 05 Part 2: Stoichiometry

(A) 

(B) 

(C) 

(D) 

## Chapter 05 Part 2: Stoichiometry

Refer to the following.

$PCl_5(g) \rightleftarrows PCl_3(g) + Cl_2(g)$

$PCl_5(g)$ decomposes into $PCl_3(g)$ and $Cl_2(g)$ according to the equation above. A pure sample of $PCl_5(g)$ is placed in a rigid, evacuated 1.00 L container. The initial pressure of the $PCl_5(g)$ is 1.00 atm. The temperature is held constant until the $PCl_5(g)$ reaches equilibrium with its decomposition products. The figures below show the initial and equilibrium conditions of the system.

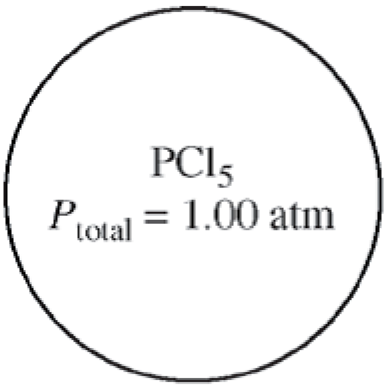



Figure 1: Initial

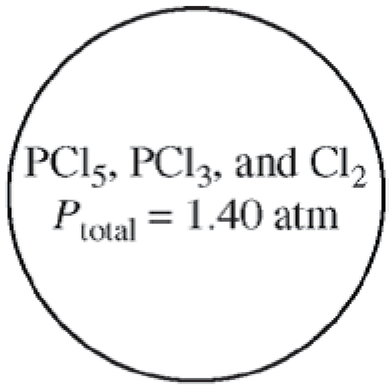



Figure 2: Equilibrium

**45.** If the decomposition reaction were to go to completion, the total pressure in the container would be

(A) 1.4 atm
(B) 2.0 atm
(C) 2.8 atm
(D) 3.0 atm

---

**46.**

$$2\ KClO_3(s) \rightarrow 2\ KCl(s) + 3\ O_2(g)$$

What is the percentage yield of $O_2$ if 12.3 g of $KClO_3$ (molar mass 123 g) is decomposed to produce 3.2 g of $O_2$ (molar mass 32 g) according to the equation above?

(A) 100%
(B) 67%
(C) 50%
(D) 33%
(E) 10%

## Chapter 05 Part 2: Stoichiometry

**47.** W($g$) + X($g$) → Y($g$) + Z($g$)

Gases W and X react in a closed, rigid vessel to form gases Y and Z according to the equation above. The initial pressure of W($g$) is 1.20 atm and that of X($g$) is 1.60 atm. No Y($g$) or Z($g$) is initially present. The experiment is carried out at constant temperature. What is the partial pressure of Z($g$) when the partial pressure of W($g$) has decreased to 1.0 atm?

(A) 0.20 atm
(B) 0.40 atm
(C) 1.0 atm
(D) 1.2 atm
(E) 1.4 atm

**48.** $NH_4NO_3$($s$) → $N_2O$($g$) + 2 $H_2O$($g$)

A 0.03 mol sample of $NH_4NO_3$($s$) is placed in a 1 L evacuated flask, which is then sealed and heated. The $NH_4NO_3$($s$) decomposes completely according to the balanced equation above. The total pressure in the flask measured at 400 K is closest to which of the following? (The value of the gas constant, $R$, is 0.082 L atm $mol^{-1}$ $K^{-1}$.)

(A) 3 atm
(B) 1 atm
(C) 0.5 atm
(D) 0.1 atm
(E) 0.03 atm

**49.**

$$CS_2(l) + 3\ O_2(g) \rightarrow CO_2(g) + 2\ SO_2(g)$$

What volume of $O_2$($g$) is required to react with excess $CS_2$($l$) to produce 4.0 L of $CO_2$($g$)? (Assume all gases are measured at 0°C and 1 atm.)

(A) 12 L
(B) 22.4 L
(C) $\frac{1}{3} \times 22.4$ L
(D) $2 \times 22.4$ L
(E) $3 \times 22.4$ L

**50.** What number of moles of $O_2$ is needed to produce 14.2 grams of $P_4O_{10}$ from P? (Molecular weight $P_4O_{10}$ = 284)

(A) 0.0500 mole
(B) 0.0625 mole
(C) 0.125 mole
(D) 0.250 mole
(E) 0.500 mole